

Web Programming

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Web programming foundations

Core network concepts

HTTP & HTTPS

Lecture: HTTP & HTTPS

Module: Core network concepts

Web protocols

- **Protocols are pretty important for web developers.**
 - ✓ **Protocols are different languages that devices use to share information.**
- There are different **protocols** like **POP**, **SMTP** and **IMAP**, that **programs use to talk about mail.**
- **File Transfer Protocol (FTP)** is used to **share files.**
- For **web developers** the **most important protocol** is **HTTP**.
 - ✓ Together with its more secure counterpart, **HTTPS**.

HTTP & HTTPS

There are two **variants** of HTTP protocol:

1. HTTP
2. HTTPS



HTTP

- HTTP stands for Hypertext Transfer Protocol.
- HTTP is **designed to transfer information** from **languages** like **HTML**.
- HTTP works by allowing someone using a **browser**, to **send requests** to **software** that handles them.
 - ✓ The **browser** is also known as a **client**.
 - ✓ The **software** that handles **requests** is also known as a **server**.

HTTPS

- **HTTPS** is an **extension** of **HTTP** that **makes** it **more secure**.
 - ✓ **HTTPS encrypts** the **requests to and from the server**.
- If **you're not using** something like **HTTPS**, it is **pretty easy to eavesdrop** the **requests to and from the server**.





HTTPS

HTTP headers

HTTP headers

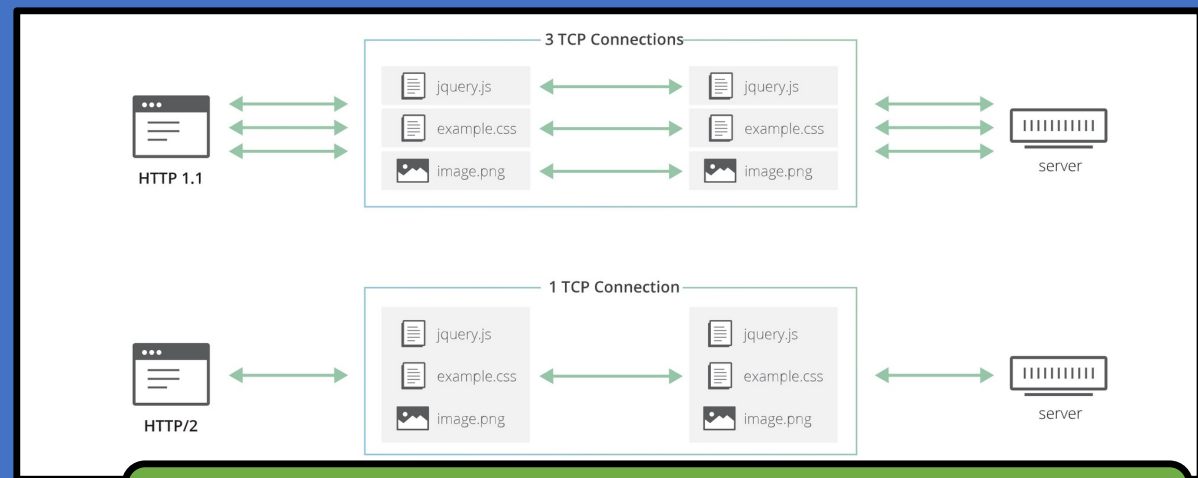
- There are **pieces of information** that **passes between client and servers**.
- One of these **pieces** are the **HTTP Headers**.
 - ✓ They provide a **lot of information** to **developers** and **hackers**.
 - ✓ **Developers** can often **read** and **manage** this information to make their **sites more secure**.
- We will take a **deep dive** into the **details** of **HTTP headers** in the **back-end development** part of this **course**.

HTTP verbs (methods)

- **HTTP** uses a series of **verbs** also known as **methods**.
 - ✓ There are **9** of these **verbs**.
 - ✓ The **most common ones** are **GET**, **POST** and **DELETE**.
- **GET** asks for a **resource** or a piece of information.
- **POST** sends some **information**.
- **DELETE** sends a **request** to **delete** something.

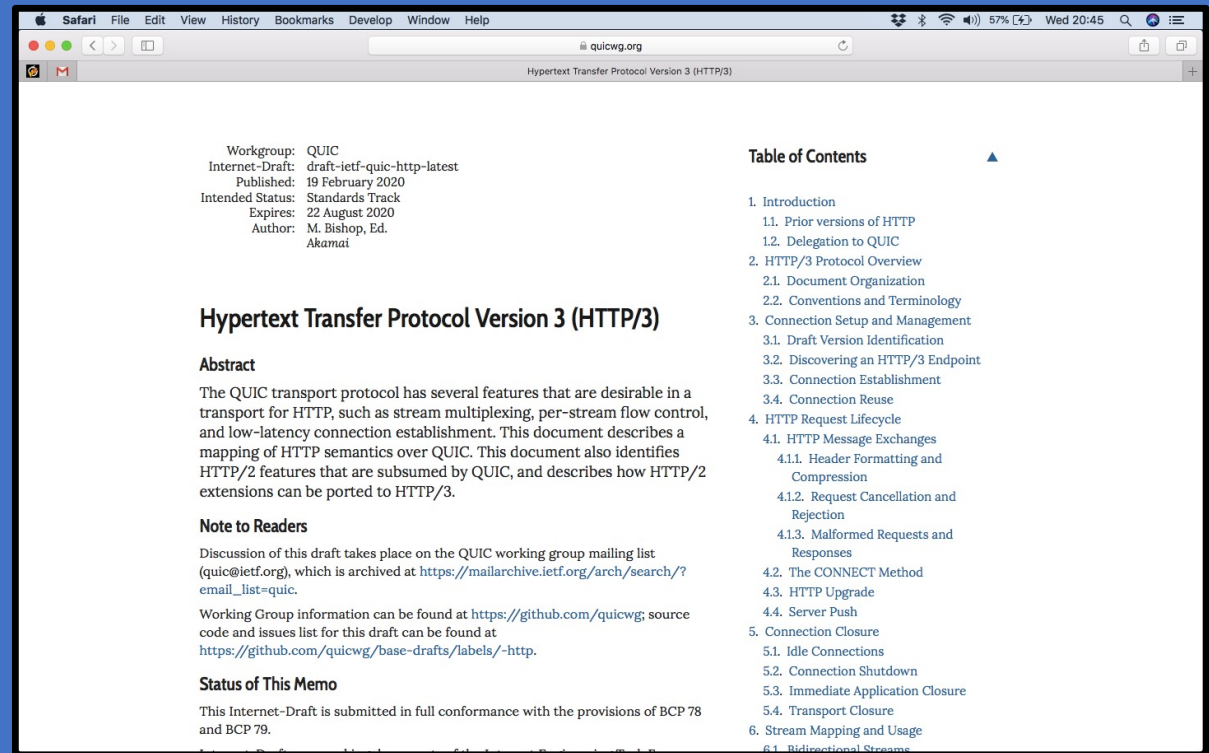
HTTP/2

- **HTTP/2** is a major revision of the **HTTP/1** protocol,
 - ✓ **HTTP/2** quickly became the standard for web traffic, and it's backwards compatible with **HTTP/1**.
- One of the main features of **HTTP/2** is **multiplexing**,
 - ✓ **Multiplexing** lets you send multiple requests at once.



Source: <https://bit.ly/39RW0tV>

HTTP continues to grow, and there are already proposals for future versions, like HTTP/3, that improve and expand the protocol.



<https://bit.ly/2T2lv3y>

TCP/IP

Lecture: TCP/IP

Module: Core network concepts

TCP/IP

- The **technology** that **makes the web possible** is a **combination of two protocols**:
 1. **TCP** – Transmission Control Protocol
 2. **IP** – Internet Protocol

TCP/IP

- Before this technology took off, most communications between computers happened by computers that were directly connected to one another.
- TCP/IP allows computers to form different types of connections.
 - ✓ This creates connections that look more like spiderwebs than straight lines.

Breaking and routing packages with TCP/IP

- TCP

- ✓ The **messages aren't** all **sent** as a **single** file, instead they're **broken down** into little **pieces** called **packets**.
- ✓ The **advantage** of the **packets** is that **data** can **travel** through the **best available paths** at any moment.

- IP

- ✓ Determines **where** the **packages** go.



IP address

- **IP** is a **series of numbers** that **describes** the **location** of the **packages**.
 - ✓ **Analogous** to the **address** of your **house**.
 - ✓ Your **ZIP code** tells the **post office** the **general location** of where you are in the **country**
 - ✓ The **street address** gets **more specific**.

192.168.0.10

200.132.100.5

Identifying resources on the web

Resources, URLs, URNs, URIs and Ports

- Each **resource** is on the *web* is **identified** by a **Uniform Resource Identifier (URI)**.
- The **identity** and the **location** of **resources** on the **Web** are mostly **given** by a single **URL (Uniform Resource Locator)**

Resources, URLs, URNs, URIs and Ports

“A URL is a locator following a universal standard that points at a specific resource somewhere on the web.”

- In the URL definition:

- ✓ Resource is the **document** you're trying to access and all its associated assets.
 - Document, photo, etc.
- ✓ Somewhere in the **world** is a **server connected** to the **Internet** whose **IP address is mapped** to the **domain** you used in a **DNS**.
 - DNS - Domain Name Server.

URLs

- Most common form of **URI**, a.k.a **web address**.

1	<code>https://developer.mozilla.org</code>
2	<code>https://developer.mozilla.org/en-US/docs/Learn/</code>
3	<code>https://developer.mozilla.org/en-US/search?q=URL</code>

- A **URL** is **composed** of different **parts**:
 - ✓ Some **mandatory** and **others** are **optional**.
- A **more complex example** might look like this:

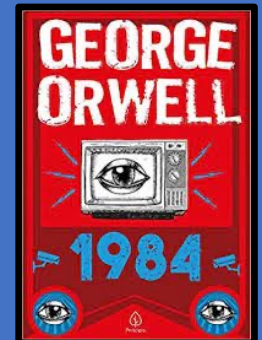
1	<code>http://www.example.com:80/path/to/myfile.html?key1=value1&key2=value2#SomewhereInTheDocument</code>
---	---

URNs

- A **Uniform Resource Name (URN)** is a **URI** that **identifies** a **resource** by **name** in a particular **namespace**.

1	urn:isbn:9780141036144
2	urn:ietf:rfc:7230

- The **two URNs** correspond to:
 - ✓ The book **1984** by **George Orwell**.
 - ✓ The **IETF specification 7230**, Hypertext Transfer Protocol (HTTP/1.1): Message Syntax and Routing.



Source: <https://bit.ly/2Tuj4Mn>

Syntax of URIs

`http://www.example.com:80/path/to/myfile.html?key1=value1&key2=value2#SomewhereInTheDocument`

Protocol: usually the **HTTP** protocol or its secured version, **HTTPS**.

Syntax of URIs

`http://www.example.com:80/path/to/myfile.html?key1=value1&key2=value2#SomewhereInTheDocument`

Domain name: authority that governs the **namespace**. It indicates which **web server** is being requested. Alternatively, it is possible to directly use an **IP address**, but because it is less convenient, it is **not often used** on the **Web**.

Syntax of URIs

`http://www.example.com:80/path/to/myfile.html?key1=value1&key2=value2#SomewhereInTheDocument`

Port number: It indicates the technical "gate" used to access the resources on the web server. It is usually omitted if the web server uses the standard ports of the HTTP protocol (80 for HTTP and 443 for HTTPS) to grant access to its resources.

Syntax of URIs

`http://www.example.com:80/path/to/myfile.html?key1=value1&key2=value2#SomewhereInTheDocument`

Path: path to the **resource** on the **web server**. In the **early days** of the **web**, a **path** like this represented a **physical file location** on the **web server**. Nowadays, it is mostly an **abstraction** handled by **web servers** without any **physical reality**.

Syntax of URIs

`http://www.example.com:80/path/to/myfile.htm?key1=value1&key2=value2#SomewhereInTheDocument`

Parameters: list of **key/value pairs** separated with the **&** symbol. The **web server** can use them to do **extra things** before **returning the resource** to the user.

Syntax of URIs

`http://www.example.com:80/path/to/myfile.html?key1=value1&key2=value2#SomewhereInTheDocument`

Anchor: anchor to **another part** of the **resource** itself.

- On an **HTML document**, the **browser** will **scroll** to the **point** where the **anchor** is **defined**;
- On a **video** or **audio document**, the **browser** will try to **go** to the **time** the **anchor** represents.

Web domain name and Domain name servers

Lecture: Web domain name and Domain name servers
Module: Core network concepts

Web domain name

Web domain name

- Determine the **address of your site** is one of the **first decisions** you'll **make** when you need to **host a site**.
 - ✓ Try to **choose a domain name** that will be **easily remembered** by **users visiting the site**.

“What's in a name? that which we call
a rose
by any other name would smell as
sweet.”

— William Shakespeare, *Romeo and Juliet*

Top-Level Domain - TLD

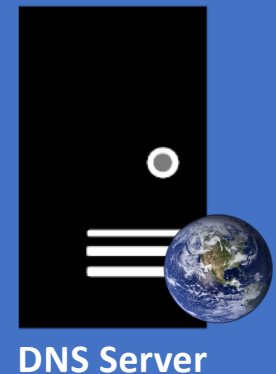
- You must **purchase a domain name** and then include a **Top-Level Domain - TLD**
 - ✓ .com, .org, .edu.
- You you can also ask **for country specific domains.**
 - ✓ .br, .ca, .fr

Top-Level Domain - TLD

- In the **old days** there were only a **few TLDs available** for **purchase**.
 - ✓ They have **recently grown** to **include** more than a **thousand**.
- **Nowadays** you can **include** things like:
 - ✓ .tech
 - ✓ .soccer
 - ✓ .space
 - ✓ ...
 - ✓ ...

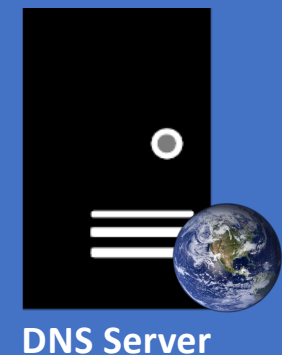
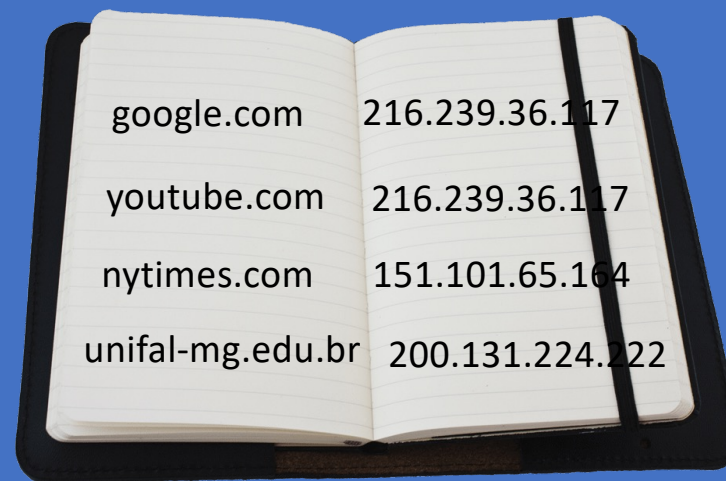
Domain Name Servers - DNS

- **Most of the time**, when we're working on the **web**, we **don't see IP addresses**.
 - ✓ But instead use **names** like **google.com** or **facebook.com**.
 - ✓ Those **names** get **translated** into **IP addresses** by a sort of **address book system** called the **Domain Name Servers (DNS)**.



DNS server

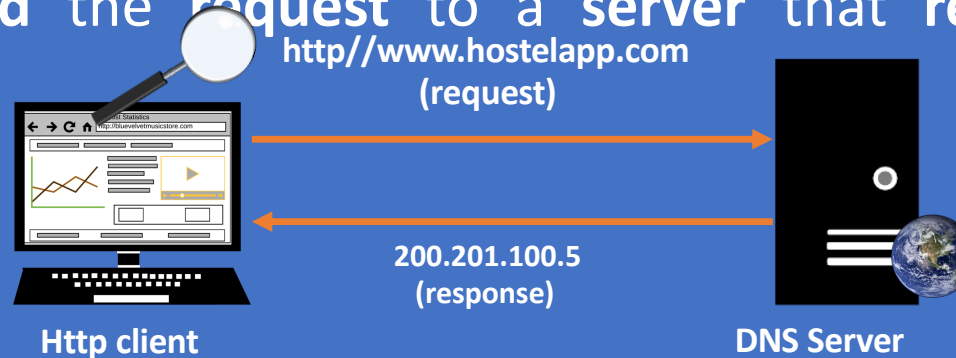
- Machine that manage **translating** the **names** that you **type** in a **browser** into **IP addresses**.
- **DNS** can be **located** by **computers** and **hackers** can sometimes **target** the **DNS servers** because it's a **way** to **overwhelm** **access** to **more** than **one website**.



DNS server

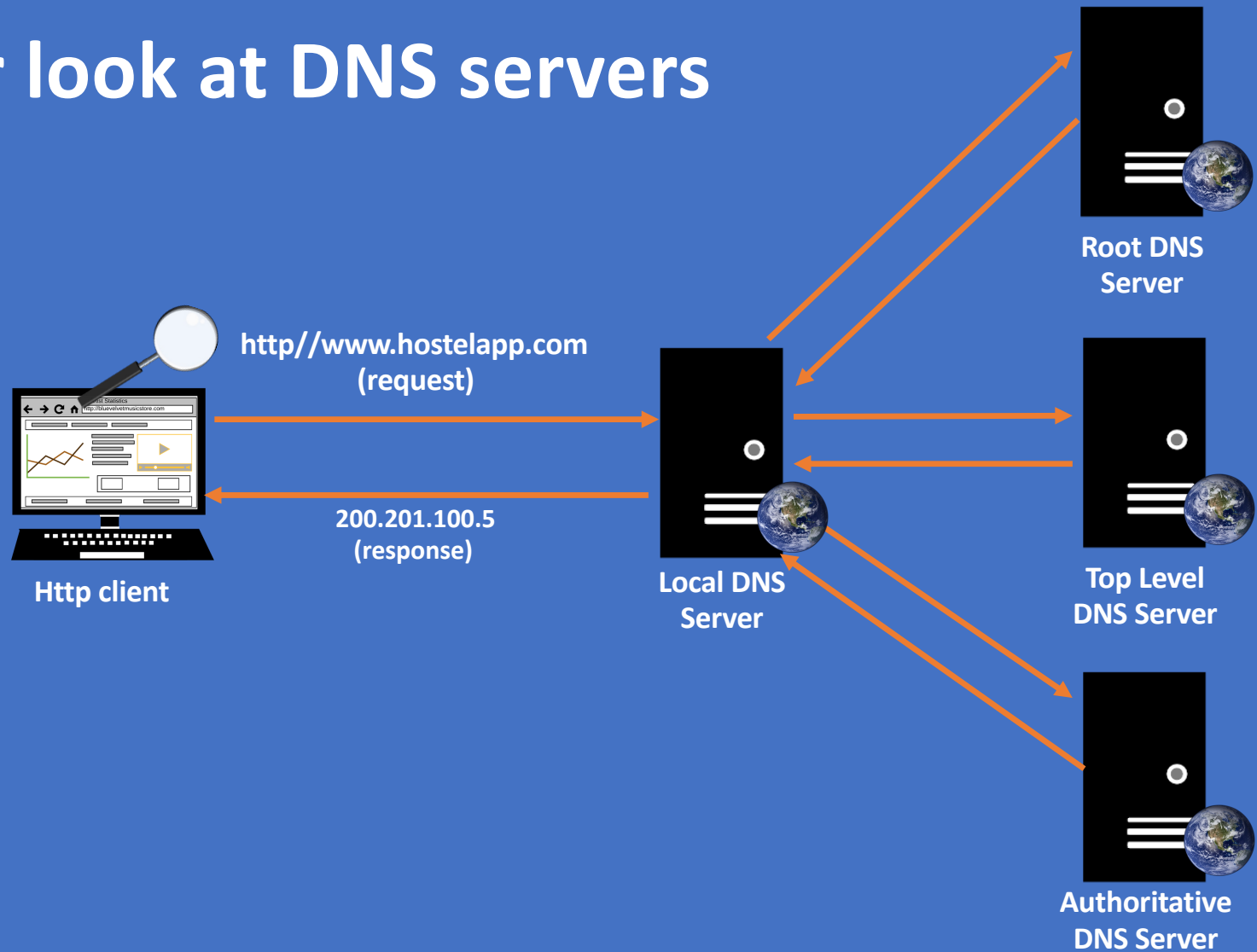
- When you type in an address like `https://facebook.com`, you're identifying that you want to use the **HTTPS protocol** and that you want to locate the **Facebook** domain using the **.com** extension.
 - ✓ More details in a subsequent lecture.

- A **DNS server** will **translate** `https://facebook.com` into a **number** and **send the request** to a **server** that **returns back a website**.

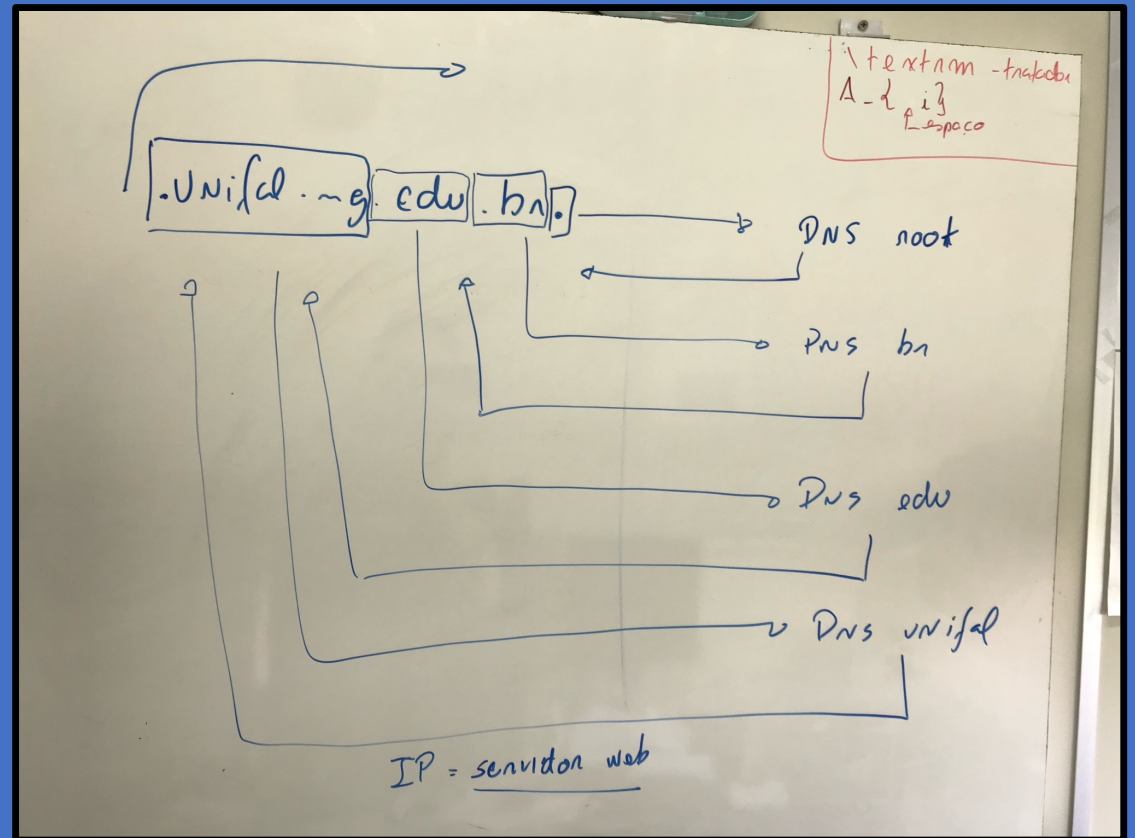


Types of DNS servers

A deeper look at DNS servers



Prof. Flavio Barbieri's "handout" about the inner working of DNS*



* Personal communication in February 2020

Questions?

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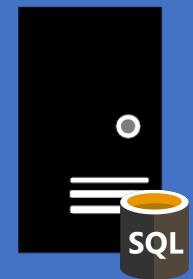
Web & Application servers

Lecture: Web & Application servers

Module: Core network concepts

Web & application servers

- One of the **most important devices** on the **web** are **servers**.
- A **server** is a **computer running specialized software**
 - ✓ A server receives requests and serves them to another device.
- **Most** of the **servers** don't even **run** on **physical machines**.
 - ✓ Servers run the software on **virtual machines** which **live** in the **cloud**.



Web servers vs. application servers

- The main difference between web server and application server resides on the type of content being served (static or dynamic):
 - ✓ A web server is meant to serve static pages
 - HTML, CSS.
 - ✓ An application server is responsible for generating dynamic content by executing server-side code.

The importance of servers for developers

- **Servers** are what **makes** the **web** possible.
- As a **back-end developer**, you will often **work** on **configuring** and **getting** the **most** out of them.
- As a **front-end developer**, you need to **know** how to **interact** and **send files** to and **from** servers.

The World Wide Web

Lecture: The World Wide Web

Module: Core network concepts

The World Wide Web

- The **WWW** is an **application** built to make information **accessible, shareable, and linkable**.
- With **HTML**, a **web document** can **link** to **contents** within itself, to **other contents** on the **same domain**, or to **any content anywhere** on the **web**.
 - ✓ That is why it's called the **World Wide Web**.

The **WWW** is literally a **web** of links tying **information** from all over the world **together** and **making** it all accessible through a **web browser**.

The World Wide Web

- The **www** was created to **allow everyone** to **share information** through **HTML documents** over the **Internet**.
 - ✓ Making a **document look pretty** and have **cool interactive features** and so on, is **secondary**.
- **What matters** is the **document** itself.



What makes the **www** so **successful** is the **fact** that **billions** of **people** across the **world** carry a **web browser** in their **pockets**.

World Wide Web

- **Many** of our **daily interactions** with **information, businesses, and services** now **happen** over the **World Wide Web**.
 - ✓ The **majority** of them happen in a **web browser** on a **computer or mobile device**.

Web document

- Every webpage is a document.
- A properly structured document is readable and accessible even without all the fancy colors and fonts and layouts.
- That's the **whole point** of the Internet:
 - ✓ Serve up the content in a uniform format, and let people decide how to access that content.
- The web browser is an advanced viewer for web documents.

From URL to the web document in the web browser

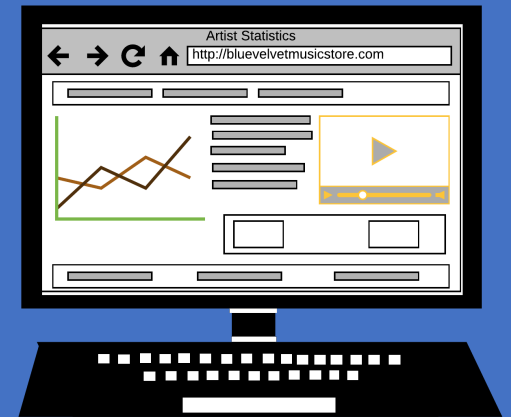
Lecture: From URL to the web document in the web browser
Module: Core network concepts

The Internet

- **Knowing** how the **Internet works** and **how** to work **with it**, all the way from the **code editor** to the **mobile browser** and beyond, is **essential** to **anyone** wanting to **publish** their **ideas** on the **web**.
- The **best place** to **start** is with something you already know, **the browser**.

What happens when you open your web browser, type in a URL in the address bar and hit Return?

If **everything** goes **well**, the **website** loads in the **browser** and you can **read**, **view** or **listen** to the content.





Content Delivery Network - CDN

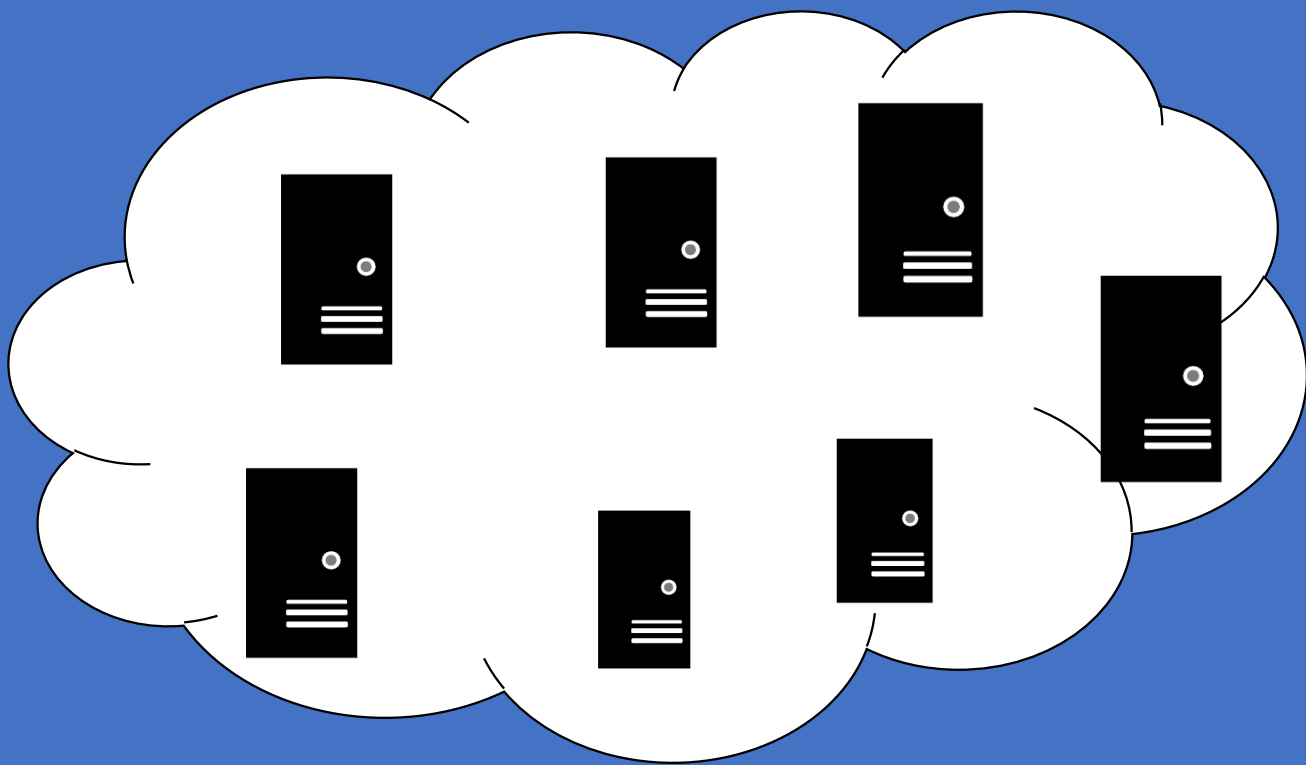
Content Delivery Network - CDN

- If your site needs to **deliver maximum speed**, it really matters how far the **customer** is from the **server**.
 - ✓ You can **place data** closer to your **users** by using a **Content Delivery Network (CDN)**.
- In a **CDN**, copies of your **site's information** are **deployed to multiple places throughout the world** so that **users** can get to it **quicker**.

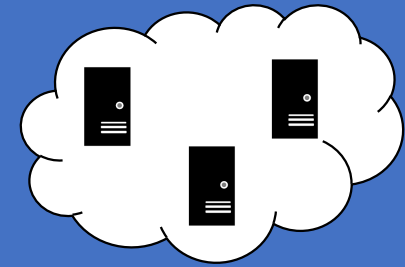
The cloud

Lecture: The cloud

Module: Core network concepts



The cloud



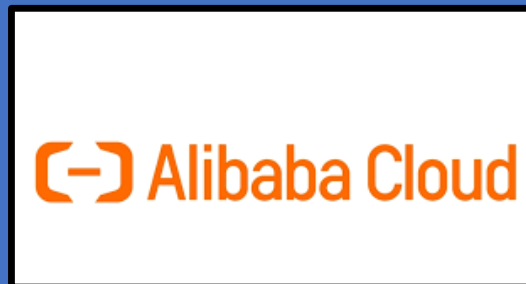
- **Worldwide network of computers that hosts applications by creating virtual machines**
 - ✓ These machines are configured, managed and run by companies like Amazon, Google or Microsoft.
- **Different server applications can live in the same machine or in different servers.**
- **Applications can specialize in business logic rather than on complex issues such as scalability and load balancing.**

Cloud services

- **Cloud services** provides **virtual machines** and **applications** that are **elastic** in nature.
 - ✓ They **grow** based on **your needs**.
 - ✓ You only **purchase** the **services** you need.
- As your **site grows** and you **gain more users**, **speed**, **memory** and **throughput** can adjust.

Cloud service providers

- Four major providers for the services:
 1. Google Cloud.
 2. Amazon Web Services.
 3. Microsoft Azure.
 4. Alibaba Cloud.



Questions?